Experiment 9

BI Security: Authentication, Authorization and Access Control using Microsoft Power BI

Aim

To understand and implement Business Intelligence security concepts including authentication, authorization, role-based access control, and data protection mechanisms using Microsoft Power BI.

Tools / Software Required

1. Microsoft Power BI Desktop
2. Microsoft Power BI Service
3. Global Superstore Dataset.xlsx
4. Microsoft Excel

Theory

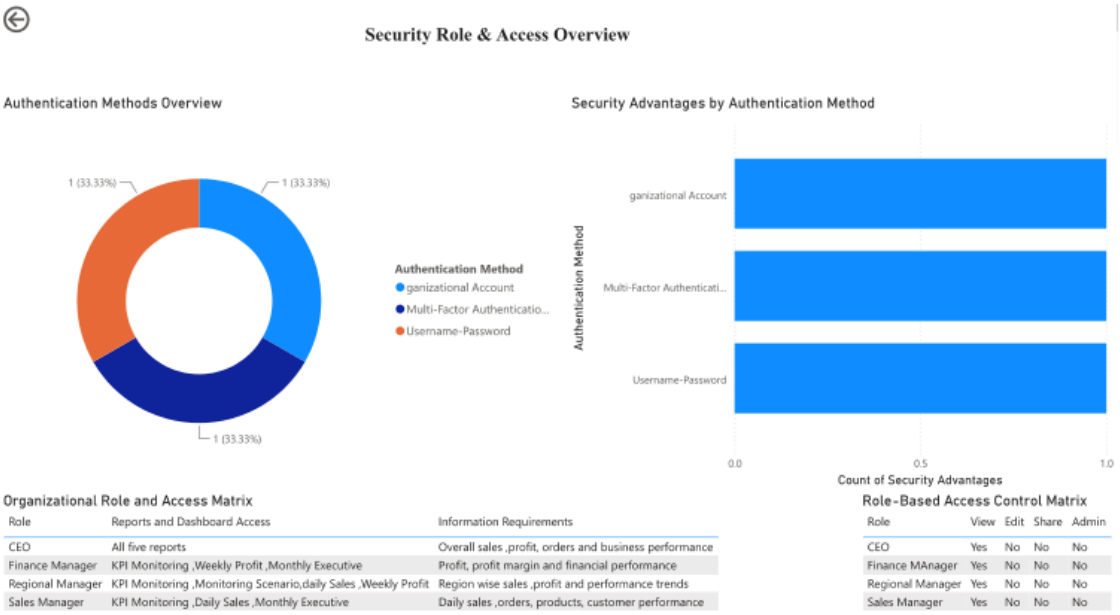
Security is a critical component of Business Intelligence systems. BI platforms often contain sensitive organizational information that must be protected from unauthorized access. Authentication verifies the identity of a user, while authorization determines the resources and actions that the user is permitted to access. Access control mechanisms ensure that users can only view data relevant to their responsibilities. Power BI supports security through workspace permissions, sharing controls, Row-Level Security (RLS), and role-based access control.

Business Scenario

Global Superstore has multiple departments including Sales, Finance, and Regional Management. Each department requires access only to information relevant to its responsibilities. Students will act as BI Security Administrators and design a secure access strategy for organizational reports and dashboards.

Conduct - Part A: User and Role Identification

Step 1: Identify organizational users such as CEO, Sales Manager, Finance Manager, and Regional Manager.
Step 2: Define information requirements for each user category.
Step 3: Create a Role Matrix showing which reports and dashboards each user should access.



Organizational Role and Access Matrix

Role	Reports and Dashboard Access	Information Requirements
CEO	All five reports	Overall sales ,profit, orders and business performance
Finance Manager	KPI Monitoring ,Weekly Profit ,Monthly Executive	Profit, profit margin and financial performance
Regional Manager	KPI Monitoring ,Monitoring Scenario,daily Sales ,Weekly Profit	Region wise sales ,profit and performance trends
Sales Manager	KPI Monitoring ,Daily Sales ,Monthly Executive	Daily sales ,orders, products, customer performance

Role-Based Access Control Matrix

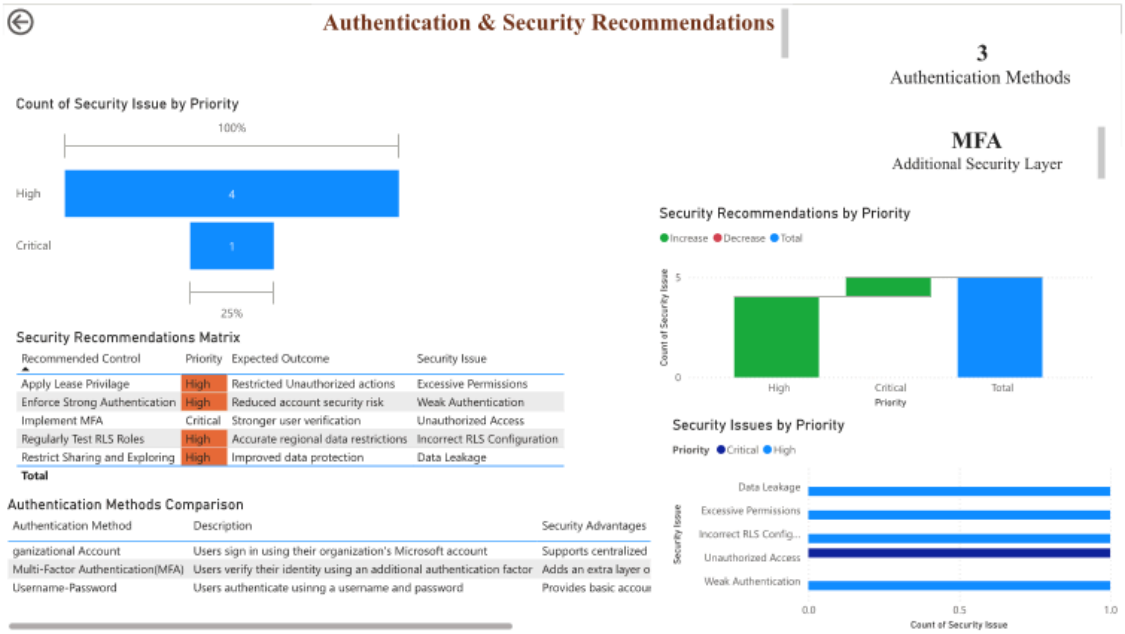
Role	View	Edit	Share	Admin
CEO	Yes	No	No	No
Finance MAnager	Yes	No	No	No
Regional Manager	Yes	No	No	No
Sales Manager	Yes	No	No	No

Conduct - Part B: Authentication Analysis

Step 4: Study authentication methods used in Power BI.
Step 5: Compare username-password authentication, organizational accounts, and multi-factor authentication.
Step 6: Document the advantages of secure authentication mechanisms.

Conduct - Part C: Authorization and Access Control Design

Step 7: Create an Access Control Matrix.
Step 8: Define permissions such as View, Edit, Share, and Admin.
Step 9: Assign permissions to different user roles.
Step 10: Justify permission assignments based on business requirements.



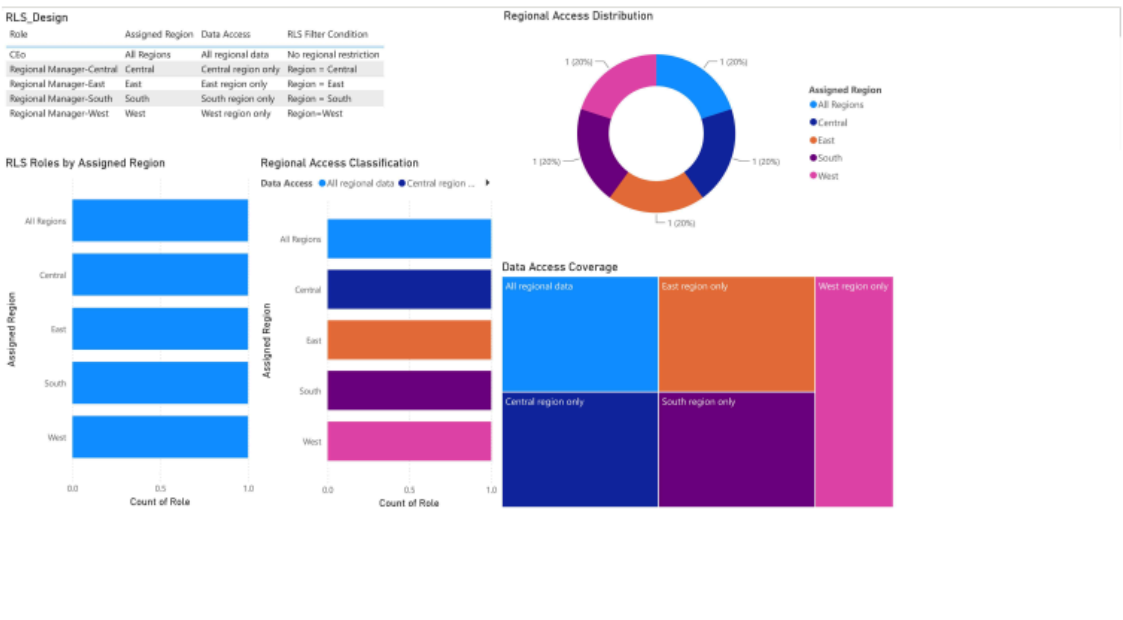
Conduct - Part D: Row-Level Security (RLS) Planning

Step 11: Design a Row-Level Security strategy.

Example:

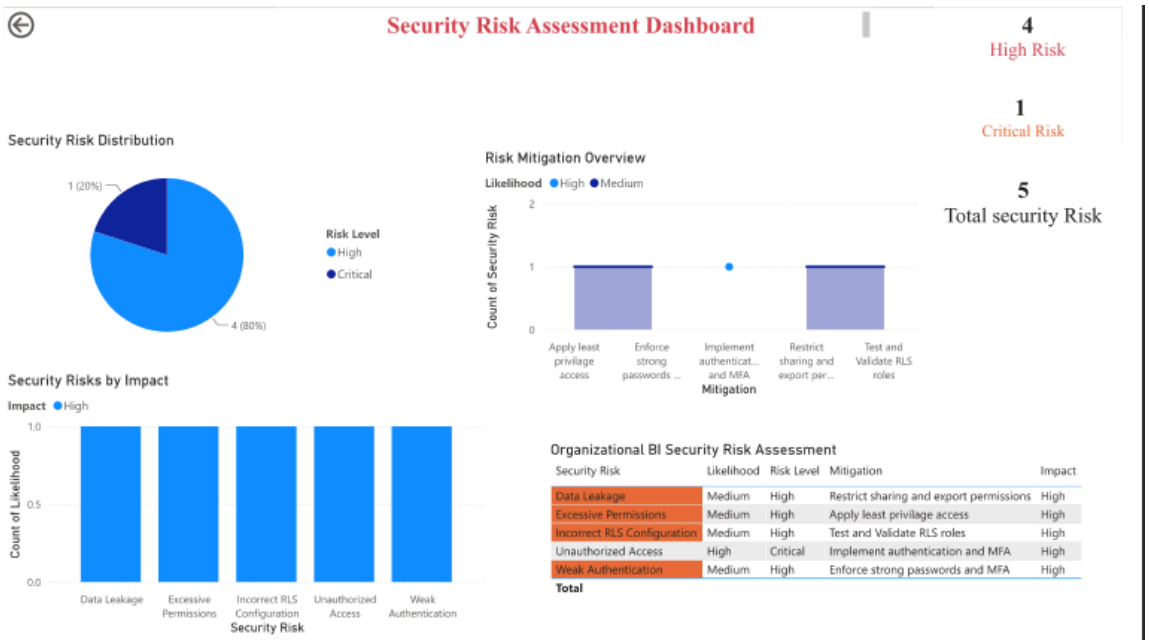
- North Region Manager can view only North Region data.
- South Region Manager can view only South Region data.

Step 12: Document how RLS improves data security.



Conduct - Part E: Security Risk Assessment

Step 13: Identify potential security risks such as unauthorized access, excessive permissions, data leakage, and weak authentication.
Step 14: Recommend security controls and mitigation measures.
Step 15: Prepare a Security Recommendation Report.



Deliverables

1. Role and Permission Matrix
2. Access Control Matrix
3. Row-Level Security Design Document
4. Security Risk Assessment Report
5. Security Recommendations Sheet

Result

A secure BI access strategy was successfully designed using authentication, authorization, and access control principles. User roles, permissions, and security measures were defined to protect organizational data and ensure controlled access.

Learning Outcomes

• Explain authentication, authorization, and access control concepts in BI systems. (CO1-BT2)
• Identify security requirements for different organizational roles. (CO1-BT2)
• Analyze access control mechanisms and permission structures. (CO4-BT4)
• Evaluate security risks associated with BI deployments. (CO4-BT4)
• Design secure access strategies for BI environments. (CO4-BT4)

Text / Reference Books

1. Sharda, Delen & Turban, Business Intelligence and Analytics.
2. Rick Sherman, Business Intelligence Guidebook.
3. Microsoft Power BI Security Documentation.
4. Successful Business Intelligence by Cindi Howson.